

mirror determines the far posterior domain in butterfly wings

Martik Chatterjee , Xin Y Yu, Noah K Brady, Gabriel C Hatto, Connor A Amendola, Robert D Reed 

Dept. of Ecology and Evolutionary Biology, Cornell University, Ithaca, United States • Dept. of Molecular Biology and Genetics, Cornell University, Ithaca, United States

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This **important** study provides evidence of a deeply conserved role for the gene *Mirror* in providing positional identity in the posterior part of butterfly and fly wings, despite increased morphological complexity of butterfly wings. The findings are **solid** for the field of evo-devo. However, the tools in butterflies are more limited than in *Drosophila* and it is more difficult to determine which specific cells are mutant and whether the effect of mutation is cell-intrinsic. The work will be of interest to evolutionary and developmental biologists working on insect wing evolution and the evolution of patterning more generally.

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Abstract

Insect wings, a key innovation that contributed to the explosive diversification of insects, are recognized for their remarkable variation and many splendid adaptations. Classical morphological work subdivides insect wings into several distinct domains along the antero-posterior (AP) axis, each of which can evolve relatively independently to produce the myriad forms we see in nature. Important insights into AP subdivision of insect wings comes from work in *Drosophila melanogaster*, however they do not fully explain the diversity of AP domains observed across broad winged insects. Here we show that the transcription factor *mirror* acts as a selector gene to differentiate a far posterior domain in the butterfly wing, classically defined as the vannus, and has effects on wing shape, scale morphology, and color pattern. Our results support models of how selector genes may facilitate evolutionarily individuation of distinct AP domains in insect wings outside of *Drosophila*, and suggest that the *D. melanogaster* wing blade has been reduced to represent only a portion of the archetypal insect wing.

Results and discussion

Insect wings vary dramatically in their size and morphology. Butterflies and moths in particular have attracted attention for their rapidly evolving wings that show extreme variation in shape and color pattern. Interestingly, the diversity of lepidopteran color patterns appears to follow

relatively simple rules, wherein an evolutionarily conserved set spot and stripe pattern elements can vary in size, color, and presence/absence along the antero-posterior (AP) axis.^{1,2} Surveys of wing pattern diversity across butterflies, considering both natural variation and genetic mutants, suggest that wings can be subdivided into five AP domains, bounded by the M1, M3, Cu2, and 2A veins as summarized in **Figure 1A**.³ Within each domain there is a strong correlation between pattern variation and wing morphology, and, conversely, between domains there is a relative degree of independence in morphological variation. We know little about how these different AP domains are differentiated during wing development, however.

Some aspects of the AP wing differentiation process have been extrapolated from *D. melanogaster* wings to butterflies. The transcription factors engrailed and spalt define AP domains in developing fly wings⁴⁻⁶ and exhibit expression patterns in butterflies that suggest some functions may be conserved. In butterflies *engrailed* transcription across the entire region posterior to the M1 vein, while its paralog *invected* marks a largely overlapping region between M1 and 2A.⁷⁻⁹ *spalt* expression is more dynamic and complex in butterflies, but appears to consistently mark a region between the R2 and M3 veins.⁹ These observations, coupled with the overall size and morphological complexity of butterfly wings, have spurred interest in the question of whether additional AP differentiation mechanisms may underlie AP domain specification in Lepidoptera and other insects.^{10,11} The identification of additional domain-determining transcription factors in broad winged insects would provide a model for the modular diversification of insect wing morphology.

The goal of this study was to identify the molecular basis of AP domain specification in the butterfly wing, posterior of the M1 engrailed boundary. We initially identified the transcription factor *mirror* as a potential candidate selector gene based on previous mRNA-seq studies,¹² coupled with *D. melanogaster* work showing that *mirror* specifies the alula – a small membranous lobe at the posterior base of the wing in some dipterans (**Figure 1C,D**).^{13,14} We used *in situ* hybridization to visualize *mirror* mRNA localization in the last-instar imaginal discs of the common buckeye butterfly, *Junonia coenia*, and observed *mirror* expression throughout the wing domain posterior to the 2A vein boundary (**Figures 2A and S1**). This expression precisely marks the vannus, or anal region (**Figures 1B, E**), of the butterfly wing – a distinct posterior domain of the wing blade populated by the anal (A) veins and bordered anteriorly by the claval fold between the cubital (Cu) and A veins in the hindwing (**Figure 2B**). As observed in various representative species across different butterfly families, this region of the hindwing is distinct from the rest of the wing; the color patterns defining the wing blade terminate at the 2A vein, and in many species it is marked by scales that differ in morphology and lack color patterns found in anterior regions of the wing (**Figures 1B and S2A**). This region has been proposed to represent the posterior-most color pattern domain in butterflies (**Figure 1A, Figure 2B**).³

To functionally assess the role of *mirror* in wing development we generated CRISPR/Cas9 mosaic knockout (mKOs) targeting the coding region of *J. coenia mirror*. We recovered many mKO mutants that consistently exhibited several distinctive phenotypes (**Table S1**), but only ever in the *mirror*-expressing posterior vannus region:

1. **Venation:** In mKO mutant hindwings we observed frequent truncation or complete loss of the loss of the 3A vein, as well as discontinuity and ectopic bifurcation of 2A veins (**hindwings: Figures 2C-G, S2, S3**; **forewings: Figure S4**).
2. **Wing shape:** mKOs consistently showed shape change along the posterior edge of the wing, especially the hindwing where the projecting lobe characteristic of the vannus was reduced and replaced by a more even, regular wing margin. The anal fold between the 2A and 3A veins, which marks the edge of the vannal lobe, was also lost in mutant clones affecting this wing region (**Figures 2C-G, S2-S3**).

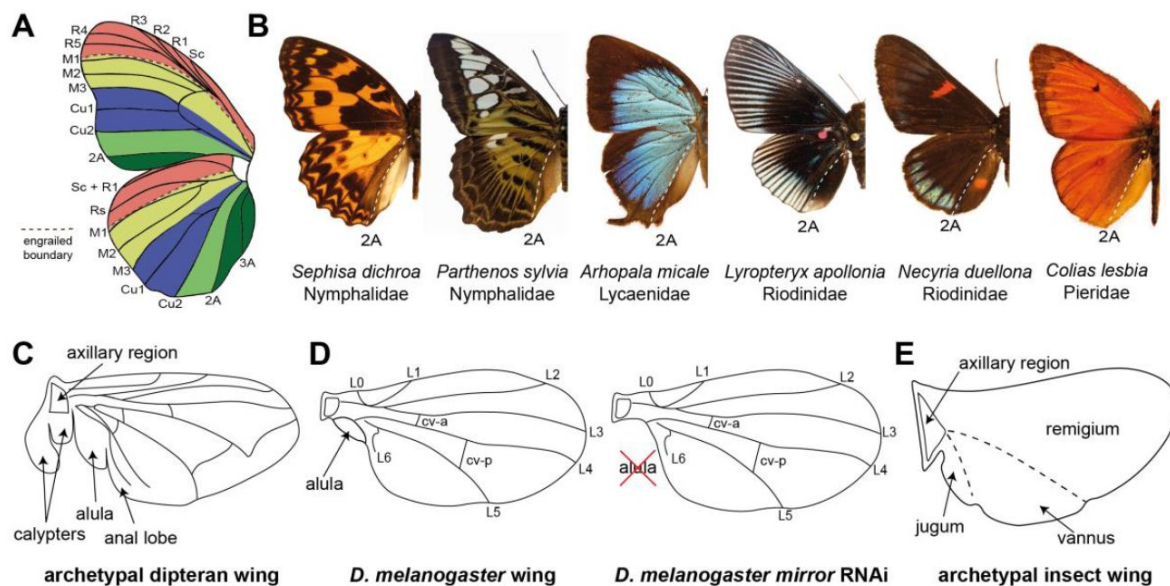


Figure 1.

Proposed AP domains of insect wings.

(A) Comparative morphology previously identified boundaries associated with color pattern variation, demarcated by the M1, M3, Cu2, and 2A veins.⁵ (B) Examples of the 2A vein marking a clear posterior color pattern boundary across multiple butterfly families. (C) Proximal posterior features of the archetypal dipteran wing include the calypters, which are only found in Calypterata clade, and the alula, a lobe at the base of the wing blade.²¹ (D) *mirror* RNAi knockdowns result in highly specific loss of the alula in *D. melanogaster*.¹³ (E) Snodgrass' model of the archetypal insect wing specifies three major domains along the anteroposterior axis: the remigium, the vannus, and the jugum. Homologies of these domains with the features of the dipteran wing (C), or butterfly wing pattern boundaries (A), has remained unclear.

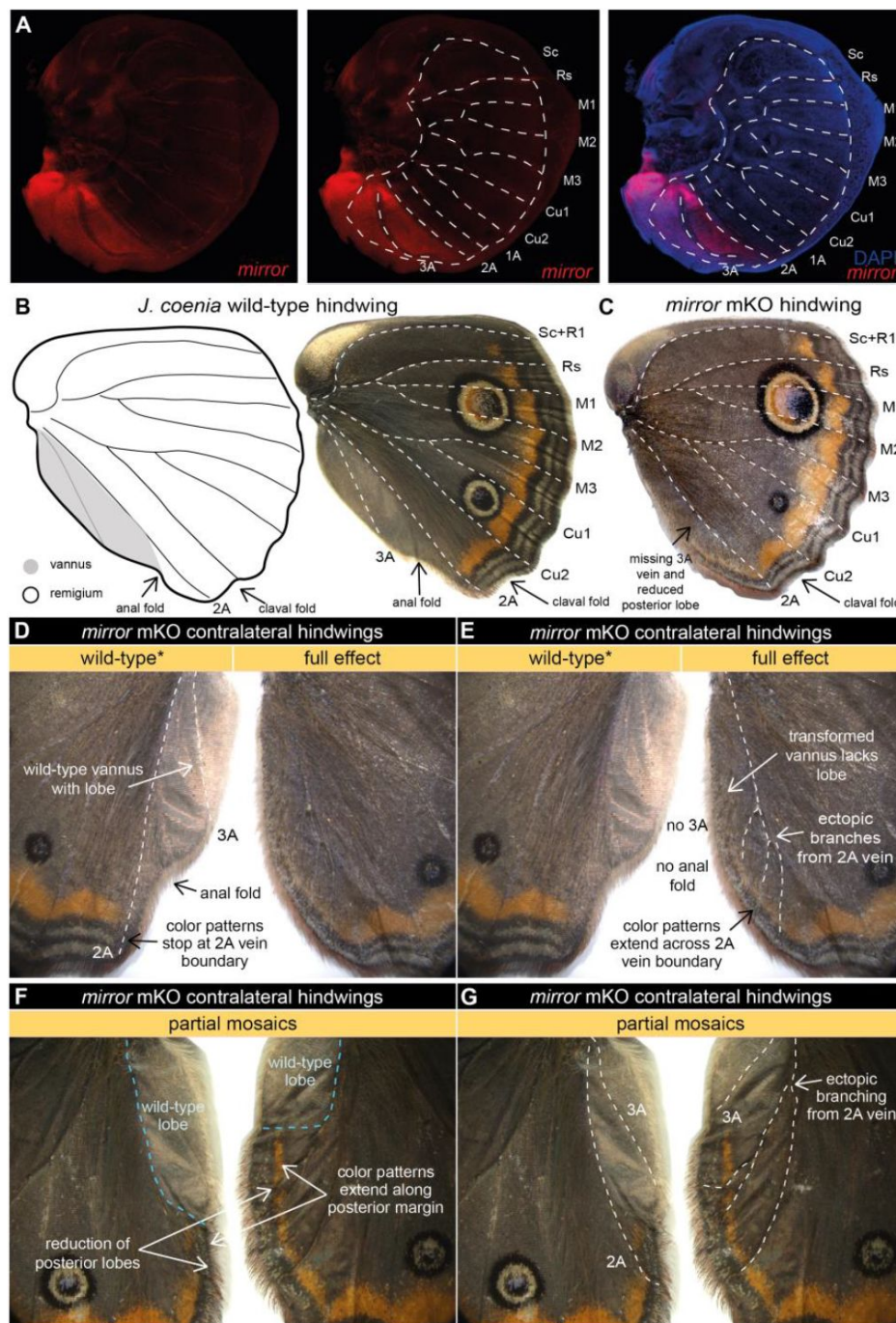


Figure 2.

***mirror* determines the identity of the far posterior vannus domain in *J. coenia*.**

(A) In *J. coenia* *mirror* (red) is expressed posterior to the 2A vein in the late last-instar hindwing imaginal discs, marking the region that will develop into the vannus. See [Figure S1](#) for forewing expression and controls. (B) In adult butterfly wings the vannus is the wing field posterior to the 2A vein, shown here in *J. coenia* as a posterior lobe with silvery scales devoid of color patterns found anterior to 2A. (C) Targeted mosaic knockouts (mKOs) of *mirror* result in loss of vannus features in *J. coenia*. (D) Annotation of wild-type features preserved on one side of a contralateral mKO. (E) Annotation of full effect mutant phenotype in the opposing wing from the individual in (D) highlights loss of lobe, loss of 3A, ectopic branching of 2A, and posterior extension of color patterns. (F) Annotation of a partial mKO highlights transformation of posterior lobe and extension of color patterns, both of which terminate at the mutant clonal boundary. (G) The individual shown in (F) also shows ectopic branching of 2A in the mutant region, but preservation of 3A in the wild-type region. For additional *mirror* mKO phenotypes see [Figures S2–S4](#).

3. **Color pattern expansion:** In mutant hindwings we consistently observed extension of wing margin color patterns across the 2A vein boundary into the posterior region of the wing. In wild-type wings the submarginal and central symmetry system color patterns terminate at the 2A vein (**dorsal: Figure 2C-G**; **ventral: Figure S3**). Remarkably, in *mirror* mKO mutants these entire color pattern systems extend past the 2A vein into the posterior regions of the wing on both dorsal and ventral wing surfaces. Specifically, submarginal bands continue past 2A to follow the posterior wing margin (**Figures 2C-G**, **S2**), while central symmetry patterns extend from 2A down to the posterior margin (**Figure S3**).
4. **Scale morphology:** In most mutants vannus scales lost their distinctive silver structural color and rounded apical edges, and transformed into brown, serrated scales typical of the remigium (**Figure S2A**). We also observed that scales along mutant wing margins often displayed elongated scales, often manifesting as a dense hair-like matt across the posterior margin (**Figure S2A**).

The knockout phenotypes we observed illustrate several things about *mirror*'s function in wing development. First, across all mKO replicates we never observed mutant phenotypes anterior to the 2A vein, which suggests that *mirror* effects are limited to the far posterior domain in which we observed *mirror* expression. Second, *mirror* knockouts appear to affect multiple aspects of wing morphology, including scale shape and coloration, overall wing shape, and wing venation. Finally, the loss of *mirror* function resulted in transformation, not loss, of the posterior domain. Even in mKO individuals that appeared to have a mutant phenotype across the entire posterior region, the *mirror*-expressing cells posterior to the 2A vein were not simply lost – the wing field continues well past the 2A vein. Instead, the identities of the post-2A scale cells changed in color and structure. In this respect, the expansion of multiple color pattern systems into the posterior domain of mutants was particularly compelling because the loss of central symmetry system and submarginal patterns in the post-2A vannus region is commonly observed in Lepidoptera (**Figure 1B**). The ability to reconstitute these complex color pattern elements in the posterior domain by knocking out a single transcription factor leads us to speculate that their patterning information may occur in a latent form that is masked by *mirror* effects. Together, our expression and knockout data suggest that *mirror* acts as a selector gene necessary to define the identity of the far posterior wing domain.

Identifying *mirror* as a vannus-specifying factor is of interest for a few reasons. First, it provides a molecular explanation for how different AP domains of the butterfly wing may be individuated to independently evolve their own shape and color pattern variations. Previous authors have proposed the existence of such individuated domains, and speculated that they may be specified by selector genes.^{3,10} Our data provide experimental support for this model, and now motivate us to identify additional factors that may specify other domain boundaries between the M1 and A2 veins.

Next, the role of *mirror* in specifying the vannus provides an important link between molecular developmental biology and Snodgrass' classical anatomical designations of the insect wing fields – i.e., the remigium, the vannus, and the jugum (**Figure 1E**).¹⁵ The remigium and the vannus are dominant features of most insect wings, while the jugum is smaller lobe-like feature that is reduced or lost in many clades. The remigium is the major anterior wing blade that is directly articulated by thoracic motor musculature to power flight, and is typically divided from the vannus by a distinct wing fold (e.g. the “anal fold” in butterflies). In many insect orders the vannus is a prominent fan-like structure, which can be a greatly enlarged gliding surface some hemimetabolous clades such as Orthoptera (crickets and grasshoppers), Dictyoptera (mantids and roaches), Phasmatodea (stick insects), and Plecoptera (stoneflies). Importantly, *mirror* knockdown in the hemimetabolous milkweed bug *Oncopeltus fasciatus* causes developmental defects in the claval and anal furrows in the posterior wing,¹⁶ which leads us to infer that *mirror*'s role in

determining the vannus may be deeply conserved in insects. This prompts us to speculate that the developmental individuation of the posterior domain by *mirror* facilitated the remarkable diversification of the vannus across the insects.

Finally, the function of *mirror* in determining the alula in *D. melanogaster*¹³ suggests that the alula may represent an evolutionarily reduced vannus. The ramifications of this are significant for reconstructing the history of the insect wing, because they suggest that the *D. melanogaster* wing blade is a lone remigium – only one-third of the archetypal insect wing (**Figure 1E**). The dipteran jugum appears to have been lost entirely, except perhaps as a membranous haltere cover (calypter) in the Calypttratae clade, as speculated by Snodgrass.¹⁵ Thus, while flies have been an important model for characterizing genes and processes that build insect wings, a more comprehensive understanding of the development and evolution of insect wings requires work in species that have wings more representative of the complete ancestral blueprint.

Materials and methods

Identity of *Iroquois* family gene *mirror* ortholog in *J. coenia*

We performed a reciprocal BLAST of amino acid sequences of *D. melanogaster Iroquois Complex* genes *araucan*, *caupolican*, and *mirror* to identify their orthologs in the latest *J. coenia* and *Heliconius erato lativitta* genome sequences on lepbasedb.org and *Tribolium castaneum* and *Apis mellifera* on NCBI. The amino acid sequences of the top hits – from *J. coenia* (JC_02265-RA, JC_02269-RA), *H. erato lativitta* (HEL_009655-RA, HEL_009656-RA), *A. mellifera* (LOC412840, LOC412839) and *T. castaneum* (LOC652944, LOC660345) were aligned to the *Iroquois* genes of *D. melanogaster* using MUSCLE.¹⁷ We used these alignments to build a maximum likelihood gene phylogeny on IQ-TREE 2¹⁸ using ModelFinder¹⁹ for estimating the most accurate substitution model for our amino acid sequences. We used FigTree to visualize the tree depicted in **Figure S5**.

HCR fluorescent *in situ* hybridization of *mirror* mRNA

This protocol was adapted and modified from Bruce et al. ([dx.doi.org/10.17504/protocols.io.bunznvf6](https://doi.org/10.17504/protocols.io.bunznvf6)). The coding sequence of JC_02269-RA was used by Molecular Instruments, Inc. (Los Angeles, CA, USA) to design a *mirror*-specific hybridization chain reaction (HCR) probe set (Lot #PRQ901). The probes are unique to the JC_02269-RA transcript as verified by BLAST search using the latest *J. coenia* genome assembly (v2) on lepbasedb.org, and do not share significant sequence similarity with any other transcripts, including the other *Iroquois* family member JC_02265-RA (see **Figure S5**). Last (5th) instar larval wing imaginal discs were dissected in cold 1x PBS. The dissected discs collected from left and right sides of the larvae were randomly assigned to control or treatment batches. Both controls and treatment batches were fixed in 1x cold fix buffer (750uL PBS, 50mM EGTA, 250uL 37% formaldehyde) on ice. The discs were then washed three times in 1x PBS with 0.1% Tween 20 (PTw) on ice spending approximately 30sec – 2min for each wash. Tissues were then gradually dehydrated by washing in 33%, 66%, and 100% MeOH (in PTw) for 2-5min / wash on ice, and stored in 100% MeOH in -20°C for days until HCR protocol was started. On the day of the HCR protocol, the wing discs were progressively rehydrated in cold 75% MeOH, 50% MeOH, and 25% MeOH in PTw spending 2-5min per wash. After rehydrating, discs were washed once for 10min and twice for 5min with PTw. The discs were then permeabilized in 300-500uL of detergent solution (1.0% SDS, 0.5% Tween, 50mM Tris-HCL pH 7.5, 1mM EDTA pH 8.0, 150mM NaCl) for 30min at room temperature. While waiting, hybridization buffer (Molecular Instruments) was warmed to 37°C (200uL/tube). After permeabilizing with detergent solution, each tube of wing discs was incubated in 200uL of pre-warmed hybridization buffer for 30min at 37°C. Probe solution was prepared by adding 0.8pmol (0.8 uL of probe from 1uM stock solution) of each probe mixture to 200uL of probe hybridization buffer at 37°C. The pre-hybridization buffer was removed, and the probe solution was added. For negative control tissues,

hybridization buffer without probes was added instead. All wing discs were incubated overnight (12-16hr) at 37°C. Before resuming the protocol, probe wash buffer (Molecular Instruments) was warmed to 37°C, amplification buffer (Molecular Instruments) was calibrated to room temperature, and a heat block was set to 95°C. The probe solution was removed and saved at -20°C to be reused. Wing discs were then washed four times in 1mL pre-warmed probe wash buffer at 37°C for 15min per wash. After the last wash step, discs were washed twice (5min/wash) with 1mL of 5x SSCT (5X sodium chloride sodium citrate, 0.1% Tween 20) at room temperature. The wing discs were pre-amplified with 1mL of equilibrated amplification buffer for 30min at room temperature. During this step, 2μL of each of hairpin h1 and 2μL of each hairpin h2 were mixed in 100μL of amplification buffer at 95°C for 90sec and cooled to room temperature in the dark for 30 min. For *mirror*-only *in situ* hybridizations, we used B1 hairpin amplifiers tagged with AlexaFluor 594 fluorophore (Molecular Instruments). For *mirror* and *wingless* double stains, we used *mirror* probes with B1 amplifier tagged with AlexaFluor 647 fluorophore and *wingless* probes (Lot #PRG129) with B3 hairpin amplifiers tagged with AlexaFluor 546 fluorophore. After 30min, the pre-amplification buffer was removed from the discs, hairpin solution added, and the tissues were then incubated in the dark at room temperature for 2-16hr. The hairpins were removed and saved at -20°C to be reused later. Excess hairpins were removed by washing 5 times (twice for 5min, twice for 30min and once for 5min) with 1mL of 5x SSCT at room temperature. The wing discs were then incubated in 50% glycerol solution (in 1X PBS) with DAPI (0.01μg/mL) overnight at 4°C. The wing discs were then mounted and visualized on a Zeiss 710 or Leica Stellaris 5 confocal microscope.

CRISPR-Cas9 mediated mutagenesis of *mirror* in *J. coenia*

Two single-guide RNAs (sgRNAs; sgRNA1-5'-GAATGGACTTGAACGGGGCA; sgRNA2-5'-AGAAACAGGGTCGATGATGA) targeting the homeobox domain of *mirror* were mixed with 500ng/μl of Cas9 nuclease and injected to *J. coenia* eggs 0.5-4hr after oviposition (n = 1042 eggs) as previously described.²⁰ Injected G0 individuals were reared on standard artificial diet, until they emerged and were immediately frozen in -20°C upon emergence. Wings of surviving G0 adults (n=99) were assayed for anomalies to detect mosaic knockout (mKO) phenotypes (n=29). Injection results and mutation phenotype frequencies are detailed in Table S1.

Validating *mirror* mutants by genotyping

DNA was extracted using E.Z.N.A. Tissue DNA Kit (Omega Bio-Tek) from the thorax of individuals that showed mutations in the wings. Extracted genomic DNA was amplified using a pair of primers flanking the sgRNA cut sites (Forward: 5'-CGCTTGTGCCACCTTAAAC, Reverse: 5'-GTATGGCTCGGGGATTCTG). Amplified DNA was run on a 2% agarose gel and were excised and purified using the MicroElute Cycle-Pure Kit (Omega Bio-Tek). Purified DNA was Sanger sequenced by Cornell Institute of Biotechnology. Example mutant alleles of *mirror* are shown in [Figure S6](#).

Supplemental information

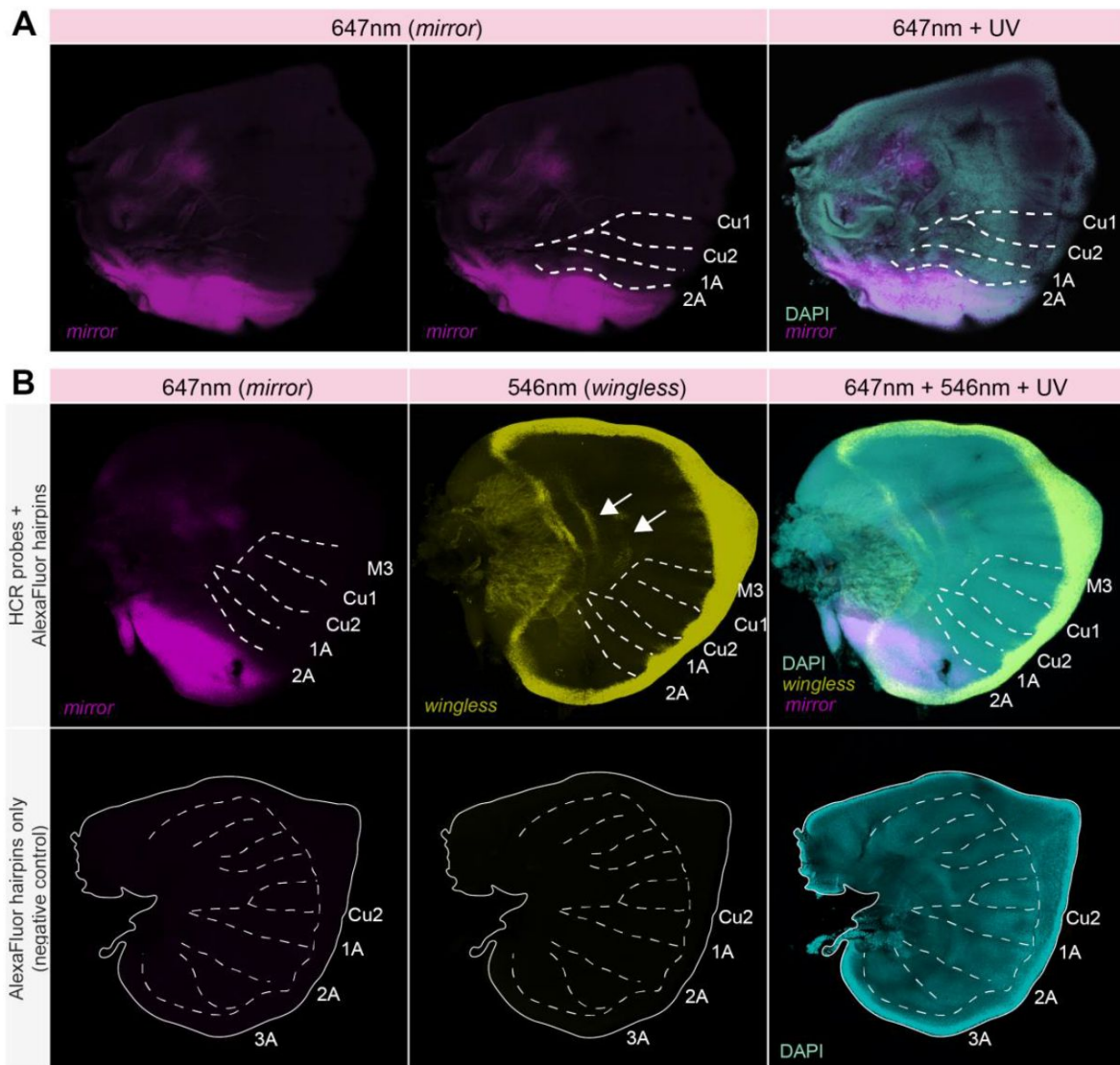


Figure S1.

HCR *in situ* hybridization of *mirror* highlight posterior domain expression in *J. coenia* wing imaginal discs. Related to Figure 2A.

(A) *mirror* is expressed posterior to the 2A vein in last-instar forewing imaginal discs of *J. coenia*. **(B)** Double *in situ* of *mirror* (Channel 1: 647nm) and *wingless* (Channel 2: 546 nm) in last instar *J. coenia* hindwing imaginal discs. Expression of *wingless* observed by HCR precisely matches expression in wing margin peripheral tissue and discal bands (white arrow) as previously described from chromogenic *in situ*⁵, thus providing a positive control for our HCR protocol. Bottom panels depict negative controls, lacking HCR probes, that show no signal in the 647nm or 546nm channels. Dashed lines show position of selected veins.



Figure S2.

mKOs of *mirror* result in partial or entire loss of posterior wing domain identity in dorsal *J. coenia* hindwings. Related to Figure 2B-E.

(A) Comparison of wild-type dorsal hindwing to mutant wings reveals complete or partial transformation of the vannus (black arrows), expansion of color patterns into the posterior wing margin (white arrows) along with vein aberrations (denoted by *), or total loss (denoted by -). The middle panels show magnified images of the posterior wing margin and vannus, with scale morphology details, in wild-type and mKO mutants (complete transformation in *JCM22* and *JCM29*; partial transformation in *JCM41*). The bottom panels show contralateral hindwing pairs from the same individuals showing asymmetry in clonal mutations affecting posterior wing margin phenotype. (B) Additional *mirror* mKO hindwings show vannus transformation phenotypes, annotated as described above.

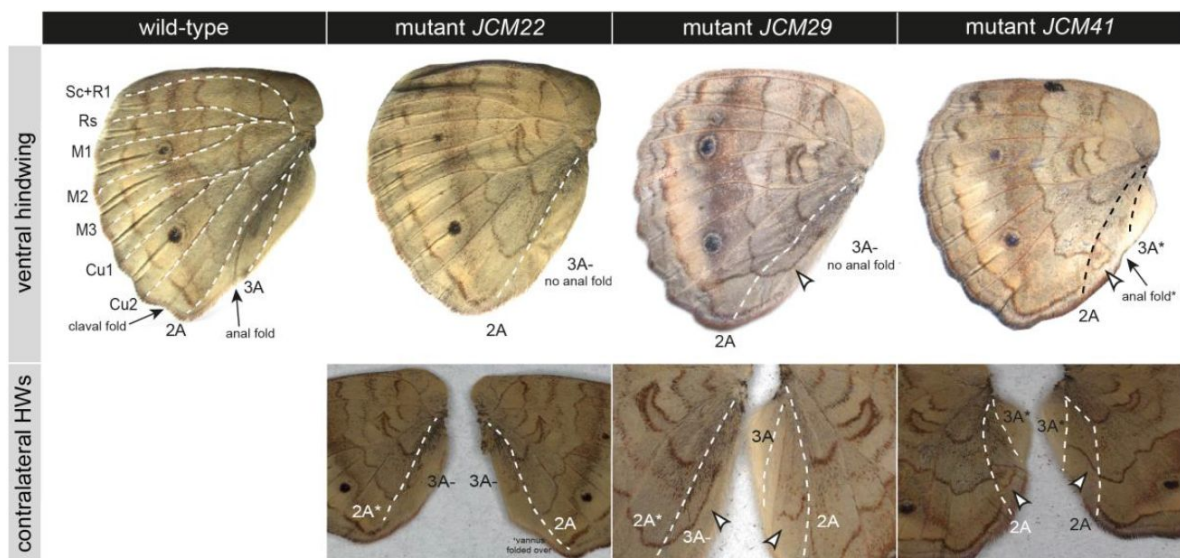


Figure S3.

Related to Figure 2B-E. mKO of *mirror* result in partial or entire transformation of posterior wing domain.

mirror mKOs in ventral *J. coenia* hindwings resulting in the expansion of color pattern elements (white arrows) as seen in asymmetric contralateral pairs from the same mutant individuals. Abnormal venation annotated by (*), missing venation annotated by (-).

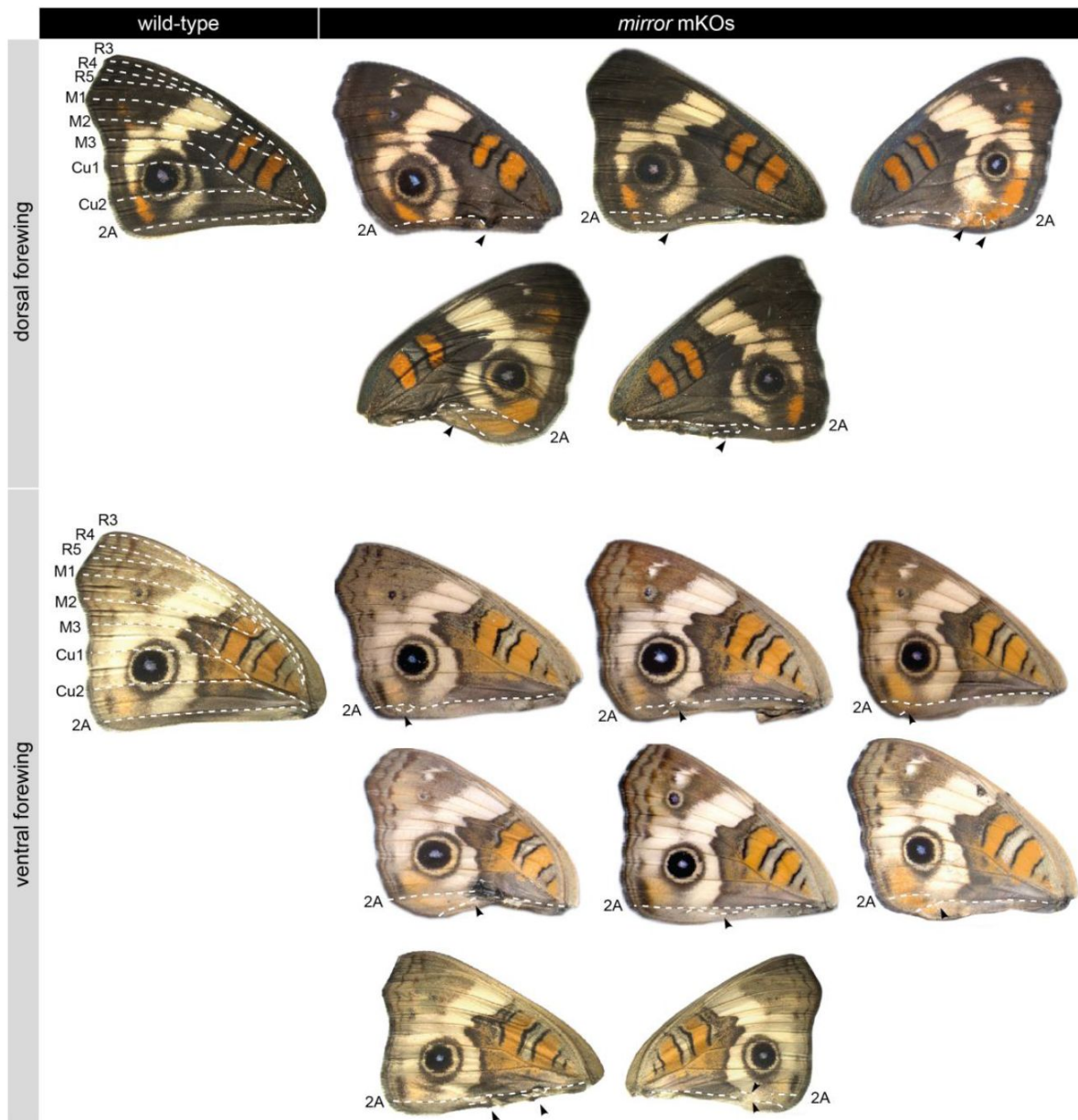


Figure S4.

***mirror* mKOs show posterior margin anomalies *J. coenia* forewings.**

mKOs of *mirror* result in anomalies along the forewing 2A vein, including vein bifurcation and ectopic vein fragments, which are often associated with color pattern disruption. Dashed lines show the position of 2A. Black arrows highlight 2A abnormalities.

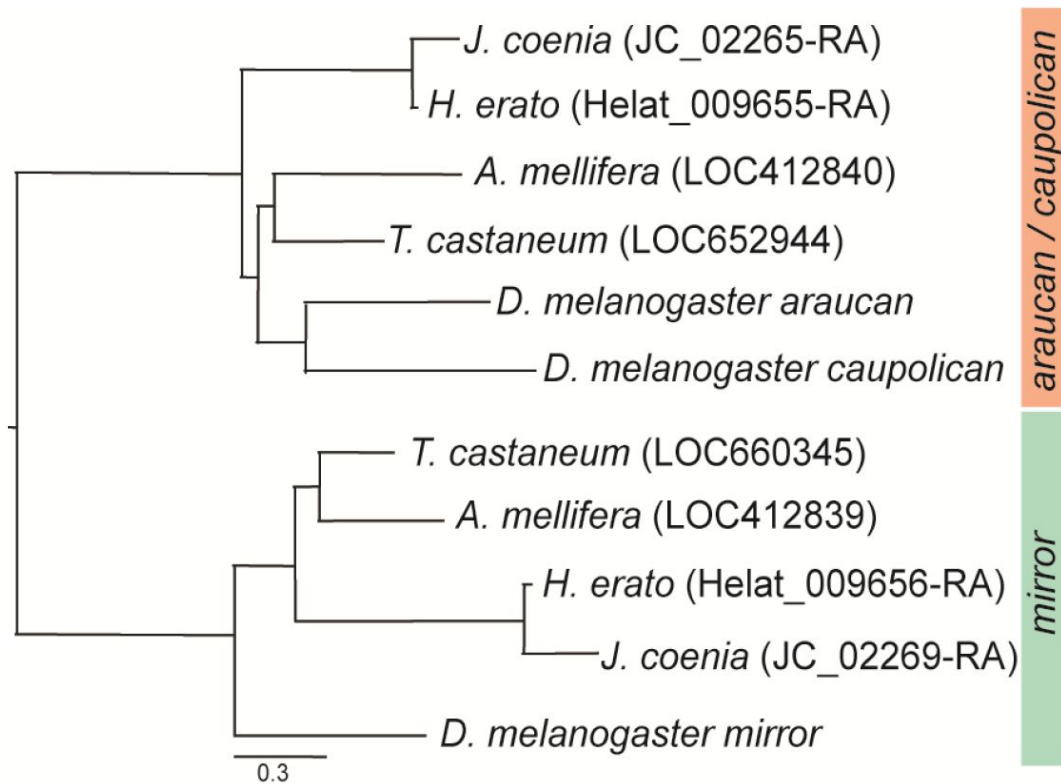


Figure S5.

Identification of *mirror* ortholog in *J. coenia*.

A maximum likelihood phylogeny of *J. coenia*, *H. erato lativitta*, *Tribolium castaneum*, *Apis mellifera*, and *D. melanogaster Iroquois Complex* genes confirms that JC_02269-RA is the ortholog of *mirror*.

Figure S6.

Sanger sequencing confirms mutations at CRISPR sgRNA site in *mirror* mKOs.

Sanger sequencing results from three different *mirror* mutant individuals. Each line denotes a different wild-type (WT) or mutant allele with indels indicated in red, and the PAM site in green.

Individuals	Indels	sg RNA1	PAM
mKO 1	WT	GAATGGACTTGAACGGGGCAAGGAGAAAAAATGCAACCAGAGAGACCACA	
	-7	GAATGGACTTGAACGG-----AGAAAAAATGCAACCAGAGAGACCACA	
	-2	GAATGGACTTGAACGGG---AAGGAGAAAAAATGCAACCAGAGAGACCACA	
	+1	GAATGGACTTGAACGG--NGCAAGGAGAAAAAATGCAACCAGAGAGACCAC	
	-2	GAATGGACTTGAACGG---CAAGGAGAAAAAATGCAACCAGAGAGACCACA	
mKO 2	WT	GAATGGACTTGAACGGGGCAAGGAGAAAAAATGCAACCAGAGAGACCACA	
	-7	GAATGGACTTGAACGG-----AGAAAAAATGCAACCAGAGAGACCACA	
	-2	GAATGGACTTGAACGG---CAAGGAGAAAAAATGCAACCAGAGAGACCACA	
	-2	GAATGGACTTGAACGGG---AAGGAGAAAAAATGCAACCAGAGAGACCACA	
	+4	GAATGGACTTGAACGGGNNNCAAGGAGAAAAAATGCAACCAGAGAGACC	
mKO 3	WT	GAATGGACTTGAACGGGGCAAGGAGAAAAAATGCAACCAGAGAGACCACA	
	-6	GAATGGACTTGAACGGG-----AGAAAAAATGCAACCAGAGAGACCACA	
	-7	GAATGGACTTGAACGG-----AGAAAAAATGCAACCAGAGAGACCACA	
	-3	GAATGGACTTGAACGG---AAGGAGAAAAAATGCAACCAGAGAGACCACA	
	-1	GAATGGACTTGAACGG--GCAAGGAGAAAAAATGCAACCAGAGAGACCACA	
	-6	GAATGGACTTGAACGG-----GAGAAAAAATGCAACCAGAGAGACCACA	
	-2	GAATGGACTTGAACGGG---AAGGAGAAAAAATGCAACCAGAGAGACCACA	

Table S1.

***mirror* CRISPR/Cas9 injection results.**

sgRNA concentration (ng/μL)	Eggs Injected	Hatched	Adults Emerged	Mutants
487 ng/μL	142	25	4	2
391 ng/μL	96	33	6	3
335 ng/μL	163	32	15	4
281 ng/μL	98	11	1	0
175 ng/μL	149	52	19	8
125 ng/μL	252	79	24	9
88.5 ng/μL	142	41	30	3
Total	1042	273	99	29

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Additional information

Author contributions

Conceptualization, M.C and R.D.R; Methodology, M.C and R.D.R; Investigation, X.Y.Y, M.C, N.K.B , G.C.H and C.A.A; Writing-Original Draft, M.C, X.Y.Y and R.D.R; Writing-Review & Editing, M.C and R.D.R; Visualization, M.C and X.Y.Y; Funding Acquisition, R.D.R and X.Y.Y; Supervision, M.C and R.D.R

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Author information

Martik Chatterjee*

Dept. of Ecology and Evolutionary Biology, Cornell University, Ithaca, United States, Dept. of Molecular Biology and Genetics, Cornell University, Ithaca, United States
ORCID iD: [0000-0003-4689-6212](https://orcid.org/0000-0003-4689-6212)

For correspondence: mc2548@cornell.edu

* Authors contributed equally

Xin Y Yu*

Dept. of Ecology and Evolutionary Biology, Cornell University, Ithaca, United States

* Authors contributed equally

Noah K Brady

Dept. of Ecology and Evolutionary Biology, Cornell University, Ithaca, United States

Gabriel C Hatto

Dept. of Ecology and Evolutionary Biology, Cornell University, Ithaca, United States

Connor A Amendola

Dept. of Ecology and Evolutionary Biology, Cornell University, Ithaca, United States

Robert D Reed

Dept. of Ecology and Evolutionary Biology, Cornell University, Ithaca, United States

For correspondence: robertreed@cornell.edu

Editors

Reviewing Editor

Michael Perry

University of California, San Diego, San Diego, United States of America

Senior Editor

Claude Desplan

New York University, New York, United States of America

Reviewer #1 (Public review):

Summary:

This short report shows that the transcription factor gene *mirror* is specifically expressed in the posterior region of the butterfly wing imaginal disk, and uses CRISPR mosaic knock-outs to show it is necessary to specify the morphological features (scales, veins, and surface) of this area.

Strengths:

The data and figures support the conclusions. The article is swiftly written and makes an interesting evolutionary comparison to the function of this gene in *Drosophila*. Based on the data presented, it can now be established that *mirror* likely has a similar selector function for posterior-wing identity in a plethora of insects.

Comments on revisions:

The revision is satisfactory. I agree with the authors that this article provides interesting insights on the evolution of insect wings. Of note, butterfly and fly wing imaginal disks differ in their mode of development: while fly wing disks grow as epithelial sacs that evaginate during metamorphosis, butterfly wing disks develop as relatively flat epithelial sheets that expand and differentiate progressively. This makes the similar role of *mirror* all the more interesting.

The revised text appropriately discuss how selector genes like *mirror* regionalize the wing during larval and pupal development. This article makes a reasonable use of CRISPR mosaic knock outs and uses contralateral controls to show the nature of the phenotypic transformations.

<https://doi.org/10.7554/eLife.96904.2.sa3>

Reviewer #2 (Public review):

This is a short and unpretentious paper. It is an interesting area and therefore, although much of this area of research was pioneered in flies, extending basic findings to butterflies would be worthwhile. Indeed, there is an intriguing observation but it is technically flawed and these flaws are far too serious to allow us to recommend publication

The authors show that *mirror* is expressed at the back of the wing in butterflies (as in flies). They present some evidence that is required for the proper development of the back of the wing in butterflies (a region dubbed the *vannus* by the ancient guru Snodgrass). But there are problems with that evidence. First, concerning the method, using CRISPR they treat embryos and the expectation is that the *mirror* gene will be damaged in groups of cell lineages, giving a mosaic animal in which some lines of cells are normal for *mirror* and others not. We do not

know where the clones or patches of cells that are defective for mirror are because they are not marked. Also, we do not know what part of the wing is wildtype and what part is mutant for mirror. When the mirror mutant cells colonise the back of the wing and that butterfly survives (many butterflies fail to develop), the back of the wing is altered in some selected butterflies. This raises a second problem: we do not know whether the rear of the wing is missing or transformed. From the images the appearance of the back of the wing is clearly different from wild type, but is that due to transformation or not? And then I believe we need to know specifically what is difference between the rear of the wing and the main part. What we see is a silvery look at the back that is not present in the main part, is it the structure of the scales? We are not told. There are other problems. Mirror is only part of a group of genes in flies and in flies both iroquois and mirror are needed to make the back of the wing, the alula (Kehl et al). What is known about iro expression in butterflies?

In flies, mirror regulates a late and local expression of dpp that seems to be responsible of making the alula. What happens in butterflies? Would a study of expression of Dpp in wildtype and mirror compromised wings be useful?

Thus, I find the paper to be disappointing for a general journal as it does little more than claim what was discovered in *Drosophila* is at least partly true in butterflies. Also it fails to explain what the authors mean by "wing domains" and "domain specification". They are not alone, butterfly workers in general appear vague about these concepts, their vagueness allowing too much loose thinking. Since these matters are at the heart of the purpose and meaning of the work reported here, we readers need a paper containing more critical thought and information. I would like to have a better and more logical introduction and discussion.

They do define what they mean by the vannus of the wing. In flies the definition of compartments is clear and abundantly demonstrated, with gene expression and requirement being limited precisely to sets of cells that display lineage boundaries. It is true that domains of gene expression in flies, for example, of the iroquois complex, which includes mirror, can only be related to pattern with difficulty. Some recap of what is known plus the opinion of the authors on how they interpret papers on possible lineage domains in butterflies might also be useful as the reader, is no wiser about what the authors might mean at the end of it!

The references are sometimes inappropriate. The discovery of the AP compartments should not be referred to Guillen et al 1995, but to Morata and Lawrence 1975.

Comments on revisions:

Nearly all the previous criticisms remain valid and are not discussed or overcome in the revision. The authors wish to draw their conclusions and we think they can do that, but they should make clear that key evidence is lacking. Thus their conclusions are speculative. But they present them more or less as facts. This is not justified. Let us suppose that clones lacking mirror do not survive or do not develop properly in the rear part of the wing and what they are seeing is occasional damage due to incomplete regeneration or to regenerative duplication?

Many clones in flies only include parts of one surface of the wing, could this happen here and how would it affect interpretations?

The null phenotype in the wing is not known but deduced from abnormal wings which "even in mKO..... appeared to have a mutant phenotype across the entire posterior region", a nice example of circular logic.

We believe the authors should be more objective and explain that their interpretations are not solid and that they should ideally be tested by finding ways of independently marking the clones. Other clonal mosaic experiments in butterflies have been done (eg <https://journals>

[.biologists.com/dev/article/150/18/dev201868/329659/Frizzled2-receives-WntA-signaling-during-butterfly](https://doi.org/10.7554/eLife.96904.2.sa2)) without cell autonomous independent markers, but they are more solid as transformed spots are made visible cell by cell by scale colour changes etc.

Their deduction that "mirror acts as a selector gene necessary to define the far posterior wing domain" is a speculative hypothesis, not a deduction and the readers should be so informed.

<https://doi.org/10.7554/eLife.96904.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Chatterjee et al., examines the role of the mirror locus in patterning butterfly wings. The authors examine the pattern of mirror expression in the common buckeye butterfly, *Junonia coenia* and then employ CRISPR mutagenesis to generate mosaic butterflies carrying clones of mirror mutant cells. They find that mirror is expressed in a well-defined posterior sector of final-instar wing discs from both hindwings and forewings and that CRISPR-injected larvae display a loss of adult wing structures presumably derived from the mirror expressing region of hindwing primordium (the case for forewings is a bit less clear since the mirror domain is narrower than in the hindwing, but there also do seem to be some anomalies in posterior regions of forewings in adults derived from CRISPR injected larvae). The authors conclude that wings of these butterflies have at least three different fundamental wing compartments, the mirror domain, a posterior domain defined by engrailed expression, and an anterior domain expressing neither mirror or engrailed. They speculate that this most posterior compartment has been reduced to a rudiment in *Drosophila* and thus has not been adequately recognized as a such a primary regional specialization.

Critique: This is a very straight-forward study and the experimental results presented support the key claims that mirror is expressed in a restricted posterior section of the wing primordium and that mosaic wings from CRISPR injected larvae display loss of adult wing structures presumably derived from cells expressing mirror (or at least nearby). The major issue I have with this paper is the strong interpretation of these findings that lead the authors to conclude that mirror is acting as a high level gene akin to engrailed in defining a separate extreme posterior wing compartment. To place this claim in context, it is important in my view to consider what is known about engrailed, for which there is ample evidence to support the claim that this gene does play a very ancestral and conserved function in a defining posterior compartments of all body segments (including the wing) across arthropods.

(A) engrailed is expressed in a broad posterior domain with a sharp anterior border in all segments of virtually all arthropods examined (broad use of a very good pan-species anti-En antibody makes this case very strong).

(B) In *Drosophila*, marked clones of wing cells (generated during larval stages) strictly obey a straight anterior-posterior border indicating that cells in these two domains do not normally intermix, thus, supporting the claim that a clear A/P lineage compartment exists.

In my opinion, mirror does not seem to be in the same category of regulator as engrailed for the following reasons:

(1) There is no evidence that I am aware of, either from the current experiments, or others that the mirror expression domain corresponds to a clonal lineage compartment. It is also unclear from the data shown in this study whether engrailed is co-expressed with mirror in

posterior most cells of *J. coenia* wing discs? If so, it does not seem justified to infer that mirror acts as an independent determinant of the region of the wing where it is expressed.

(2) The mirror is not only expressed in a posterior region of the wing in flies but also in the ventral region of the eye. In *Drosophila*, mirror mutants not only lack the alula (derived approximately from cells where mirror is expressed), but also lacks tissue derived from the ventral region of the eye disc (although this ventral tissue loss phenotype may extend beyond the cells expressing mirror).

In summary, it seems most reasonable to me to think of mirror as a transcription factor that provides important development information for a diverse set of cells in which it can be expressed (posterior wing cells and ventral eye cells) but not that it acts as a high level regulator as engrailed.

Recommendation:

While the data provided in this succinct study are solid and interesting, it is not clear to me that these findings support the major claim that mirror defines an extreme posterior compartment akin to that specified by engrailed. Minimally, the authors should address the points outlined above in their discussion section and greatly tone down their conclusion regarding mirror being a conserved selector-like gene dedicated to establishing posterior-most fates of the wing. They also should cite and discuss the original study in *Drosophila* describing the mirror expression pattern in the embryo and eye and the corresponding eye phenotype of mirror mutants: McNeill et al., *Genes & Dev.* 1997. 11: 1073-1082; doi:10.1101/gad.11.8.1073.

<https://doi.org/10.7554/eLife.96904.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This short report shows that the transcription factor gene mirror is specifically expressed in the posterior region of the butterfly wing imaginal disk, and uses CRISPR mosaic knock-outs to show it is necessary to specify the morphological features (scales, veins, and surface) of this area.

Strengths:

*The data and figures support the conclusions. The article is swiftly written and makes an interesting evolutionary comparison to the function of this gene in *Drosophila*. Based on the data presented, it can now be established that mirror likely has a similar selector function for posterior-wing identity in a plethora of insects.*

We thank the reviewer for their feedback.

Weaknesses:

This first version has minor terminological issues regarding the use of the terms "domains" and "compartment".

We acknowledge that the terminologies “domains” and “compartments” might lead to confusion. To avoid confusion we have removed the term “compartment” from the manuscript.

Reviewer #2 (Public Review):

This is a short and unpretentious paper. It is an interesting area and therefore, although much of this area of research was pioneered in flies, extending basic findings to butterflies would be worthwhile. Indeed, there is an intriguing observation but it is technically flawed and these flaws are serious.

The authors show that mirror is expressed at the back of the wing in butterflies (as in flies). They present some evidence that is required for the proper development of the back of the wing in butterflies (a region dubbed the vannus by the ancient guru Snodgrass). But there are problems with that evidence. First, concerning the method, using CRISP they treat embryos and the expectation is that the mirror gene will be damaged in groups of cell lineages, giving a mosaic animal in which some lines of cells are normal for mirror and others are not. We do not know where the clones or patches of cells that are defective for mirror are because they are not marked. Also, we do not know what part of the wing is wild type and what part is mutant for mirror. When the mirror mutant cells colonise the back of the wing and that butterfly survives (many butterflies fail to develop), the back of the wing is altered in some selected butterflies. This raises a second problem: we do not know whether the rear of the wing is missing or transformed. From the images, the appearance of the back of the wing is clearly different from the wild type, but is that due to transformation or not? And then I believe we need to know specifically what the difference is between the rear of the wing and the main part. What we see is a silvery look at the back that is not present in the main part, is it the structure of the scales? We are not told.

Thank you for this feedback. We appreciate that many readers may not be accustomed to looking at mosaic knockouts. As discussed in a previous review article (Zhang & Reed 2017), we rely on a combination of contralateral asymmetry and replicates to infer mutant phenotypes. For many genes (e.g. pigmentation enzymes) mutant clones are obvious, but for other types of genes (e.g. ligands) clone boundaries are sometimes not directly diagnosable. It is simply a limitation of our study system. Nonetheless, you see for yourself that “the back of the wing is altered in some butterflies” – the effects of deleting *mirror* are clear and repeatable.

In terms of interpreting mutant phenotypes, we agree that that paper would benefit from a better description of the specific effects. Therefore, we have included an improved, more systematic description of phenotypes, along with better-annotated figures showing changes in wing shape and venation, scale coloration, and color pattern transformation (e.g. posterior elongation of the orange marginal stripes).

There are other problems. Mirror is only part of a group of genes in flies and in flies both iroquois and mirror are needed to make the back of the wing, the alula (Kehl et al). What is known about iro expression in butterflies?

In *Drosophila* *mirror*, *araucan*, and *caupolican* comprise the so-called *Iroquois Complex* of genes. As denoted in Figure S4 and in Kerner et al (doi: <https://doi.org/10.1186/1471-2148-9-74>) the divergence of *araucan* and *caupolican* into two separate paralogs is restricted to *Drosophila*. As in most insects, butterflies have only two *Iroquois Complex* genes: *araucan* and *mirror*. We tested the role of *araucan* in *Junonia coenia* as shown in our pre-print: <https://doi.org/10.1101/2023.11.21.568172>. Its expression appears to be restricted to early pupal wings

where it is transcribed in all scale-forming cells. Mosaic *araucan* KOs resulted in a change in scale iridescent coloration associated with changes in the laminar thickness of scale cells.

In flies, mirror regulates a late and local expression of dpp that seems to be responsible for making the alula. What happens in butterflies? Would a study of the expression of Dpp in wildtype and mirror compromised wings be useful?

We thank the reviewer for the proposal and agree that a future study comparing Dpp in wild-type versus *mirror* KO butterflies would be useful to clarify the mechanism of Dpp signalling in wing development. It is not clear, however, that the results of a Dpp experiment would change the conclusions of our current study therefore we decided not to undertake these additional experiments for our revision.

Thus, I find the paper to be disappointing for a general journal as it does little more than claim what was discovered in Drosophila is at least partly true in butterflies.

We respect that the reviewer does not have a strong interest in the comparative aspects of this study. Fair enough. This report is primarily aimed at biologists interested in the evolutionary history of insect wings.

Also, it fails to explain what the authors mean by "wing domains" and "domain specification". They are not alone, butterfly workers, in general, appear vague about these concepts, their vagueness allowing too much loose thinking.

A domain is “a region distinctively marked by some physical feature”. This term is used extensively in the developmental biology literature (e.g. “expression domain”, “embryonic domain”, “tissue domain”, “domain specification”) and is found throughout popular textbooks (e.g. Alberts et al. “The Cell”, Gilbert “Developmental Biology”). We prefer the term “domain” because of its association in the *Drosophila* literature with transcription factors that define fields of cells. We specifically avoided using the term “compartment” because of its association with cell lineage, which we have not tested.

Since these matters are at the heart of the purpose and meaning of the work reported here, we readers need a paper containing more critical thought and information. I would like to have a better and more logical introduction and discussion.

We would like the very same thing, of course, and we hope the reviewer finds our revised manuscript to be more satisfying to read.

The authors do define what they mean by the vannus of the wing. In flies the definition of compartments is clear and abundantly demonstrated, with gene expression and requirement being limited precisely to sets of cells that display lineage boundaries. It is true that domains of gene expression in flies, for example of the iroquois complex, which includes mirror, can only be related to patterns with difficulty. Some recap of what is known plus the opinion of the authors on how they interpret papers on possible lineage domains in butterflies might also be useful as the reader, is no wiser about what the authors might mean at the end of it!

We thank the reviewer for this suggestion. However, our experiments have little to contribute to the topic of cell lineage compartmentalization. We have therefore opted to avoid speculating on this topic to prevent confusion and to keep the manuscript focused on our experimental results.

The references are sometimes inappropriate. The discovery of the AP compartments should not be referred to Guillen et al 1995, but to Morata and Lawrence 1975. Proofreading is required.

We thank the reviewer for suggesting this important reference. We have included it in our revision.

Reviewer #3 (Public Review):

Summary:

*The manuscript by Chatterjee et al. examines the role of the mirror locus in patterning butterfly wings. The authors examine the pattern of mirror expression in the common buckeye butterfly, *Junonia coenia*, and then employ CRISPR mutagenesis to generate mosaic butterflies carrying clones of mirror mutant cells. They find that mirror is expressed in a well-defined posterior sector of final-instar wing discs from both hindwings and forewings and that CRISPR-injected larvae display a loss of adult wing structures presumably derived from the mirror expressing region of hindwing primordium (the case for forewings is a bit less clear since the mirror domain is narrower than in the hindwing, but there also do seem to be some anomalies in posterior regions of forewings in adults derived from CRISPR injected larvae). The authors conclude that the wings of these butterflies have at least three different fundamental wing compartments, the mirror domain, a posterior domain defined by engrailed expression, and an anterior domain expressing neither mirror nor engrailed. They speculate that this most posterior compartment has been reduced to a rudiment in *Drosophila* and thus has not been adequately recognized as such a primary regional specialization.*

Critique:

This is a very straightforward study and the experimental results presented support the key claims that mirror is expressed in a restricted posterior section of the wing primordium and that mosaic wings from CRISPR-injected larvae display loss of adult wing structures presumably derived from cells expressing mirror (or at least nearby). The major issue I have with this paper is the strong interpretation of these findings that lead the authors to conclude that mirror is acting as a high-level gene akin to engrailed in defining a separate extreme posterior wing compartment. To place this claim in context, it is important in my view to consider what is known about engrailed, for which there is ample evidence to support the claim that this gene does play a very ancestral and conserved function in defining posterior compartments of all body segments (including the wing) across arthropods.

(1) Engrailed is expressed in a broad posterior domain with a sharp anterior border in all segments of virtually all arthropods examined (broad use of a very good panspecies anti-En antibody makes this case very strong).

*(2) In *Drosophila*, marked clones of wing cells (generated during larval stages) strictly obey a straight anterior-posterior border indicating that cells in these two domains do not normally intermix, thus, supporting the claim that a clear A/P lineage compartment exists.*

In my opinion, mirror does not seem to be in the same category of regulator as engrailed for the following reasons:

(1) There is no evidence that I am aware of, either from the current experiments, or others that the mirror expression domain corresponds to a clonal lineage compartment.

It is also unclear from the data shown in this study whether engrailed is co-expressed with mirror in the posterior-most cells of J. coenia wing discs. If so, it does not seem justified to infer that mirror acts as an independent determinant of the region of the wing where it is expressed.

(2) Mirror is not only expressed in a posterior region of the wing in flies but also in the ventral region of the eye. In Drosophila, mirror mutants not only lack the alula (derived approximately from cells where mirror is expressed), but also lack tissue derived from the ventral region of the eye disc (although this ventral tissue loss phenotype may extend beyond the cells expressing mirror).

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Recommendation:

While the data provided in this succinct study are solid and interesting, it is not clear to me that these findings support the major claim that mirror defines an extreme posterior compartment akin to that specified by engrailed. Minimally, the authors should address the points outlined above in their discussion section and greatly tone down their conclusion regarding mirror being a conserved selector-like gene dedicated to establishing posterior-most fates of the wing. They also should cite and discuss the original study in Drosophila describing the mirror expression pattern in the embryo and eye and the corresponding eye phenotype of mirror mutants: McNeill et al., Genes & Dev. 1997. 11: 1073-1082; doi:10.1101/gad.11.8.1073.

We thank the reviewer for their summary, critique, and recommendations. We agree with everything the reviewer says. Honestly, however, we were surprised by these comments because we took great care in the paper to never refer to *mirror* as a compartmentalization gene or claim it has a function in cell lineage compartmentalization like engrailed. As pointed out, we lack clonal analyses to test for compartmentalization. This is why we used the term “domain” instead of “compartment” in the title and throughout the manuscript. Nevertheless, we have recrafted the discussion in the manuscript, including completely removing the term “compartment”, to better avoid implications that *mirror* plays a role in cell lineage compartmentalization.

We also thank the reviewer for recommending the paper about the role of *mirror* in eye development. For the sake of keeping the paper focused, however, we decided not to broach the topic of *mirror* functions outside the context of wing development.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

I have minor comments for improvement.

The abstract and introductions are terminologically problematic when they refer to the concept of compartment and compartment boundaries. Allegedly this confusion has previously propagated in several articles related to butterfly wing development, which keeps alienating this literature from being taken seriously by fly specialists, for example. So it is important to use the right terms. I will try to explain point by point here, but I would appreciate it if the authors could undertake a significant rewrite taking these comments into account. The authors use the terms compartment and compartment boundary. This has a very specific use in developmental genetics: mitotic clones never cross a boundary (or compartment). I think the authors can keep referring to the

equivalent of the A-P boundary, which is situated somewhere between M1-M2 based on unpublished data from the Patel Lab, and is not very well defined (Engrailed expression moves a little bit during development in this area). Domain is a looser term and can be used more liberally to describe genetically defined regions.

- "Classical morphological work subdivides insect wings into several distinct domains along the antero-posterior (AP) axis, each of which can evolve relatively independently." Yes. This concept of domain and individuation seems important. You could make a proposed link to selector genes here.

- "There has been little molecular evidence, however, for AP subdivision beyond a single compartment boundary described from *Drosophila melanogaster*." Incorrect, and this conflates "domain" and "compartment".

Flies have wing AP domains too, that pattern their veins (see the cited Banerjee et al).

- "Our results confirm that insect wings can have more than one posterior developmental domain, and support models of how selector genes may facilitate evolutionarily individuation of distinct AP domains in insect wings". Yes, and I like the second part of the sentence. Still, I would recommend simply deleting "confirm that insect wings can have more than one posterior developmental domain, and" because this is neglecting previous work on AP genetic regionalization in both flies (vein literature) and butterflies (e.g. McKenna and Nijhout, Banerjee et al).

- "Analyses of wing pattern diversity across butterflies, considering both natural variation and genetic mutants, suggest that wings can be subdivided into at least five AP domains, bounded by the M1, M3, Cu2, and 2A veins respectively, within each of which there are strong correlations in color pattern variation and wing morphology (Figure 1A)". Yes, and I would recommend emphasizing they correspond to welldefined gene expression domains as mentioned in Banerjee et al, or McKenna and Nijhout.

- "The anterior-most of these domains, bordered by the M1 vein, appears to correspond to an AP compartment boundary originally described by cell lineage tracing in *Drosophila melanogaster*, and later supported in butterfly wings by expression of the Engrailed transcription factor. Interestingly, however, *D. melanogaster* work has yet to reveal clear evidence for additional AP domain boundaries in the wing." Confusingly, because the first sentence is about compartments while the second is about AP domains. I also think the claim that Dmel has no other known AP domains is dubious because Spalt is highly regionalized in flies.

- "Previous authors have proposed the existence of such individuated domains, and speculated that they may be specified by selector genes.^{5,10} Our data provide experimental support for this model, and now motivate us to identify factors that specify other domain boundaries between the M1 and A2 veins." Yes, I completely agree with this way to emphasize the selector effect, and to link it to the concept of "individuated domain"

We cannot thank the reviewer enough for the time and thought they devoted to giving helpful suggestions to improve our manuscript. We have applied all of the above recommendations to the revision.

Fig. S1: the field needs to move away from Red/Green microscopy images, for accessibility reasons.

The easiest fix here would be to change the red channels to magenta.

Green/Magenta provides excellent contrast and accessibility in general in 2-channel images.

We thank the reviewer for this suggestion. We have improved the color accessibility of Fig. S1.

<https://doi.org/10.7554/eLife.96904.2.sa0>